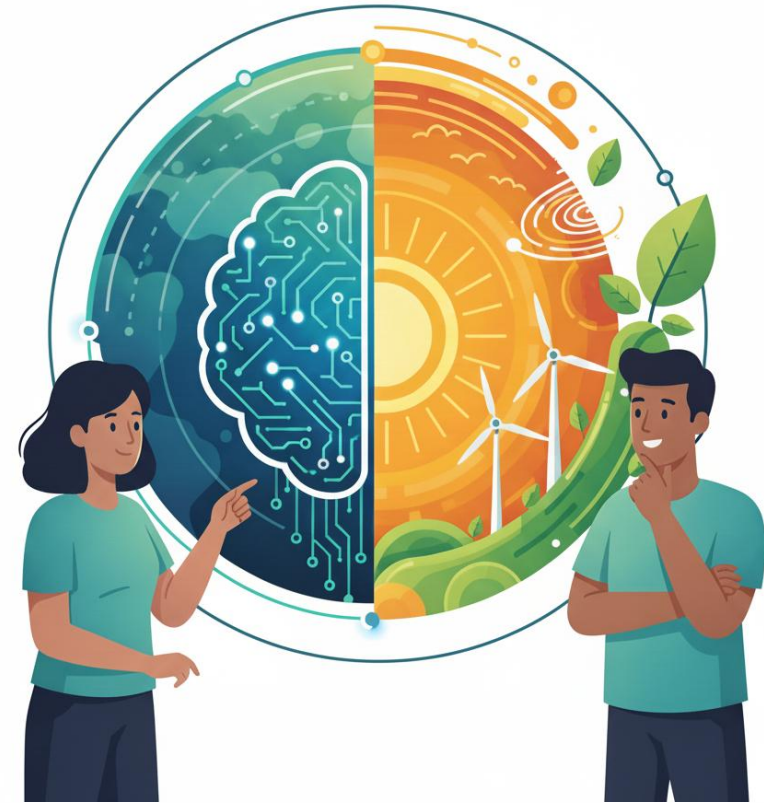


AI Adoption Training

Workshop#1: Introduction to LLMs and
Context Engineering

Spring 2026



Introductions

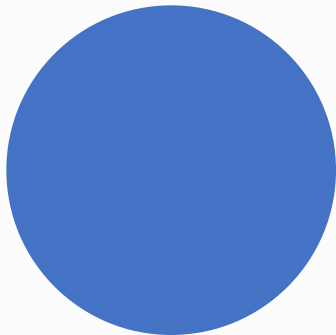


Sean Mullin:

- Works at the intersection of economics, technology and public policy.
- Currently Senior Fellow, Munk School of Global Affairs & Public Policy – focused on economics of AI, AI adoption, and sovereign AI policy.
 - Latest: [Sovereign by Design: Strategic Options for Canadian AI Sovereignty](#)
- Previously, served as Special Advisor, Economic Affairs to the Prime Minister (2023-25), founded the Brookfield Institute for Innovation + Entrepreneurship (2015–2022), and served as Senior Advisor and Director of Policy to the Premier of Ontario (2007–2011).
- Degrees in business, economics & computer science (Oxford, McGill, U of T).

Today's Agenda

1. **Introductions** (5 mins)
2. **Context and Background Information** (10 mins)
3. **Part 1 – LLMs 101: Foundations & Key Concepts** (45 mins)
4. **Break** (2 mins)
5. **Part 2 – Prompt and Context Engineering** (45 mins)
6. **Wrap Up** (5 mins)



Overview of Training Program

Team Training Overview:

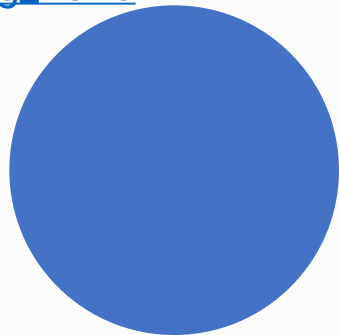
- Five workshops, scheduled over the next two months.
- Minimum of two sessions for each workshop.
- First session focuses on understanding how LLMs work (2hrs), followed by four more topic-specific sessions (90 mins each)
- Recommend that everyone join the first session and then can sign up for subsequent sessions based on interest.

Workshops Overview:

- **Workshop 1: Intro to LLMs & Context Engineering (TODAY)**
- **Workshop 2: AI for Writing, Analysis & Communications**
- **Workshop 3: AI for Operational Excellence**
- **Workshop 4: AI for Advanced Research Techniques**
- **Workshop 5: Advanced AI Topics — Coding, Models & Agents**

Materials available at: https://github.com/mullinsean/ai_training_2026

Follow along and experiment using Claude, ChatGPT, Gemini or your tool of choice.



Workshop Schedule – Spring 2026

#	Workshop	Duration	Session A	Session B	Session C
1	LLMs & Context Engineering	2 hours	Tue, Apr 28 11:00 AM – 1:00 PM	Thu, Apr 30 1:00 PM – 3:00 PM	Mon, May 4 10:00 AM – 12:00 PM
2	Writing, Comms & Presentations	90 min	Tue, May 5 11:00 AM – 12:30 PM	Wed, May 15 1:30 PM – 3:00 PM	-
3	Analysis & Operations	90 min	Tue, May 19 11:00 AM – 12:30 PM	Thu, May 21 1:30 PM – 3:00 PM	-
4	Advanced Research Techniques	90 min	Wed, May 27 1:30 PM – 3:00 PM	Tue, Jun 2 11:00 AM – 12:30 PM	-
5	Coding Tools, Models & Agents	90 min	Tue, Jun 16 11:00 AM – 12:30 PM	Thu, Jun 18 1:30 PM – 3:00 PM	-

All times Eastern • For each workshop, participants can join whichever session is most convenient.

Context: “AI is Everywhere”

Latest AI tech show strong promise of being a new General-Purpose Technology (GPT)

- AI is everywhere, including increasingly showing up in the economic statistics.
- While we may be in a speculative boom, the underlying technology is very real:
 - Frontier models (eg: ChatGPT 5.5, Claude 4.7, Gemini 3.1) are achieving or surpassing human-level performance across a wide variety of benchmarks.
 - Even if progress halted with today’s models, we’d likely have 5+ years of productivity gains ahead of us. And models continue to get better.
- The organizations (and countries!) that learn how to adopt this technology will have tremendous economic advantages.
- Like previous technological cycles, there will be disruption and risks along the way. We need to address these, while embracing AI adoption across the economy.

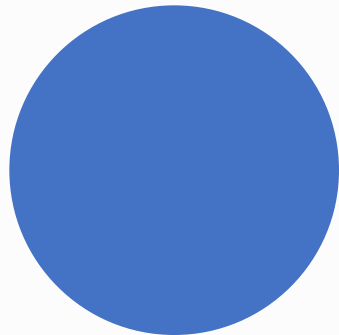


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Learning Approach

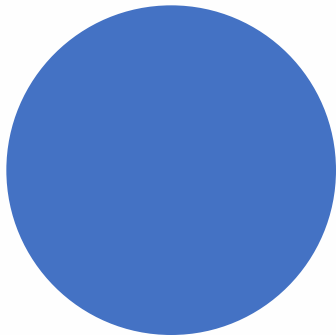
Start with unpacking LLMs and then move into more specific use cases.

- The key technology that is underlying recent “explosive” gains in AI performance are Large Language Models (LLMs), which are based off the Transformer architecture
 - ChatGPT, Claude, Gemini, DeepSeek are all LLMs; but these LLMs also power new AI tools like MS Copilot, Perplexity.ai, Cursor, etc.
- Understanding how LLMs work under the hood can generate *general insights* and techniques for using AI across all of these tools.
- We will focus on both the strengths and the limitations of LLM tools.



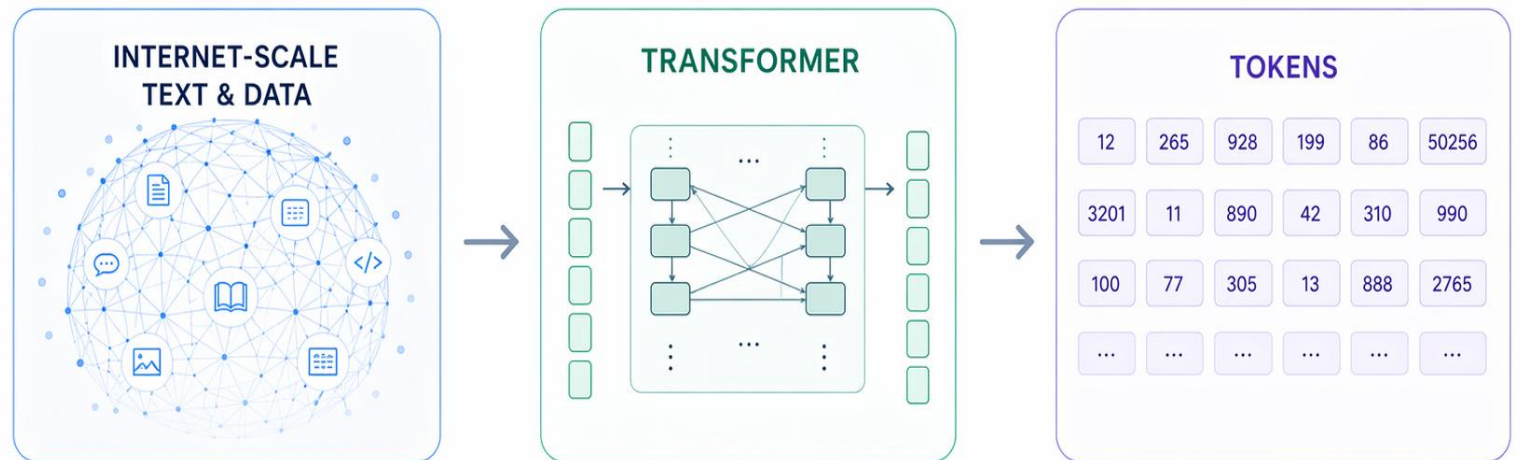


LLMs 101: Foundations & Key Concepts



LLMs 101: Modern AI is Built on LLMs

- Large Language Models (LLMs) are the core underlying technology powering modern AI tools
- They are enormously complex mathematical models with literally billions of parameters. They have been *trained on data*, not written like tradition software.
- LLMs are **stochastic**, which means they determine *probabilistically* the most likely next token.
- Tokens are the key unit of measurement for LLMs: usage limits, context window constraints, API billing are all measured in tokens.



ANTHROPIC



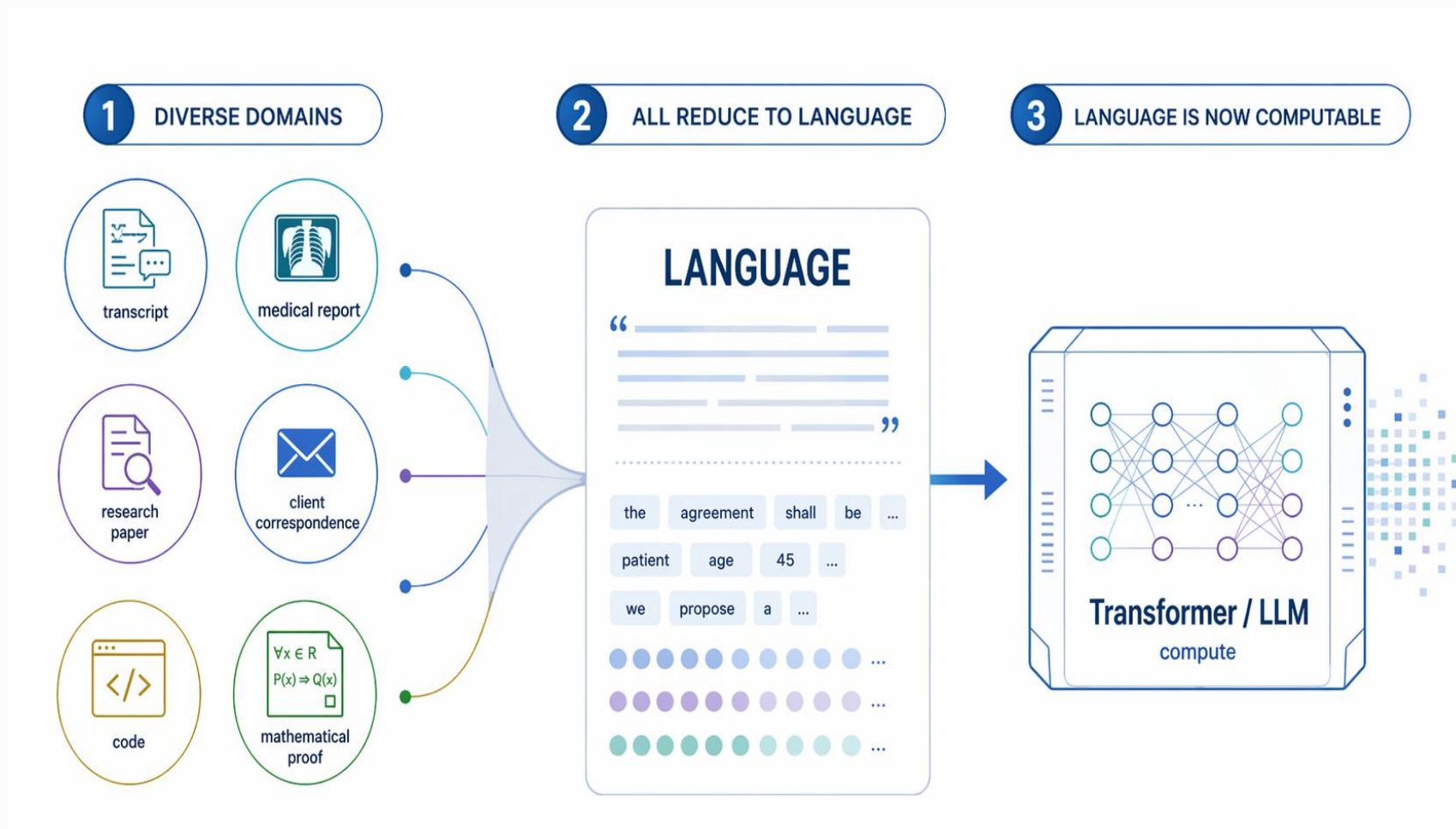
xAI

Meta



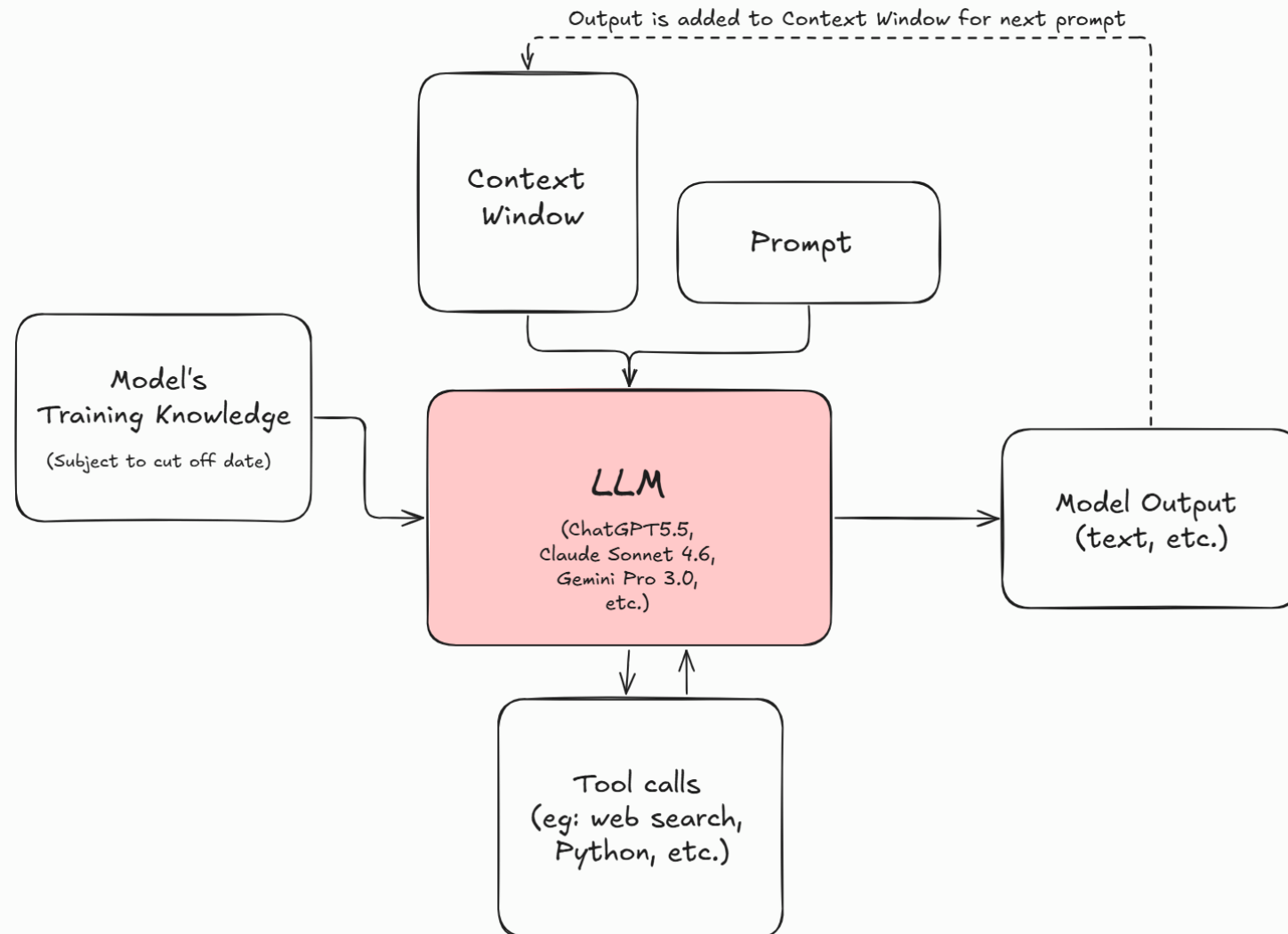
LLMs: A New General Purpose Technology

Anything expressible in language is now within reach of compute



- Prior AI was narrow: classifying images, playing Go, recommending films
- LLMs cross domains without retraining: the first cognitive technology that is **general**
- Economists call technologies that reshape every sector General Purpose Technologies (eg: steam engine, electricity, semiconductors, etc.)

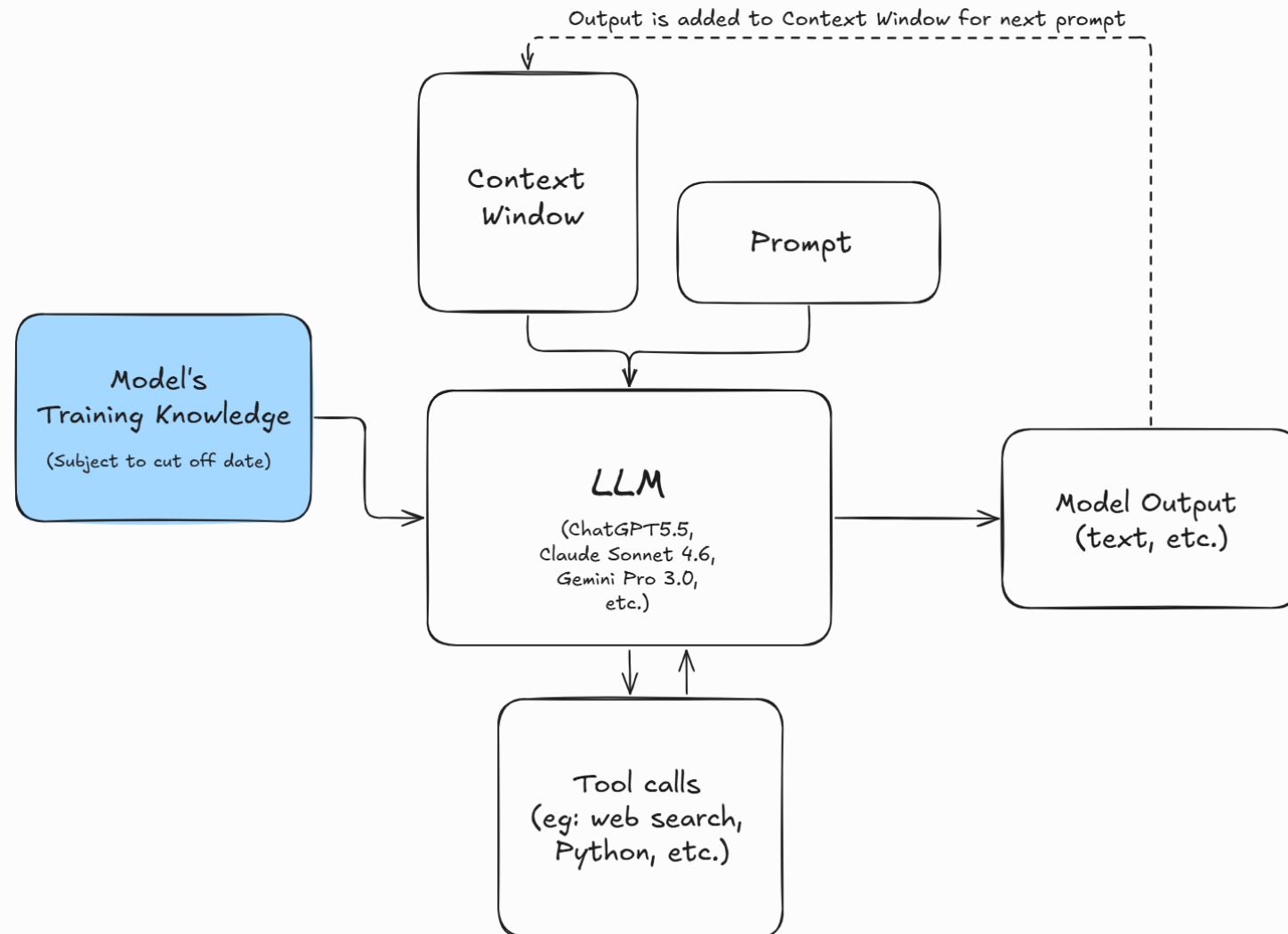
How LLMs Work



LLMs:

- An LLM combines information from the model's training knowledge, the context window, tool calls and the user's prompt to generate output.
- Under the hood, LLM interfaces use different models for different purposes:
 - ChatGPT "Instant" vs "Thinking", Claude Sonnet vs Opus
- Frontier models are frequently being updated, with major upgrades being released every 6-12 months.

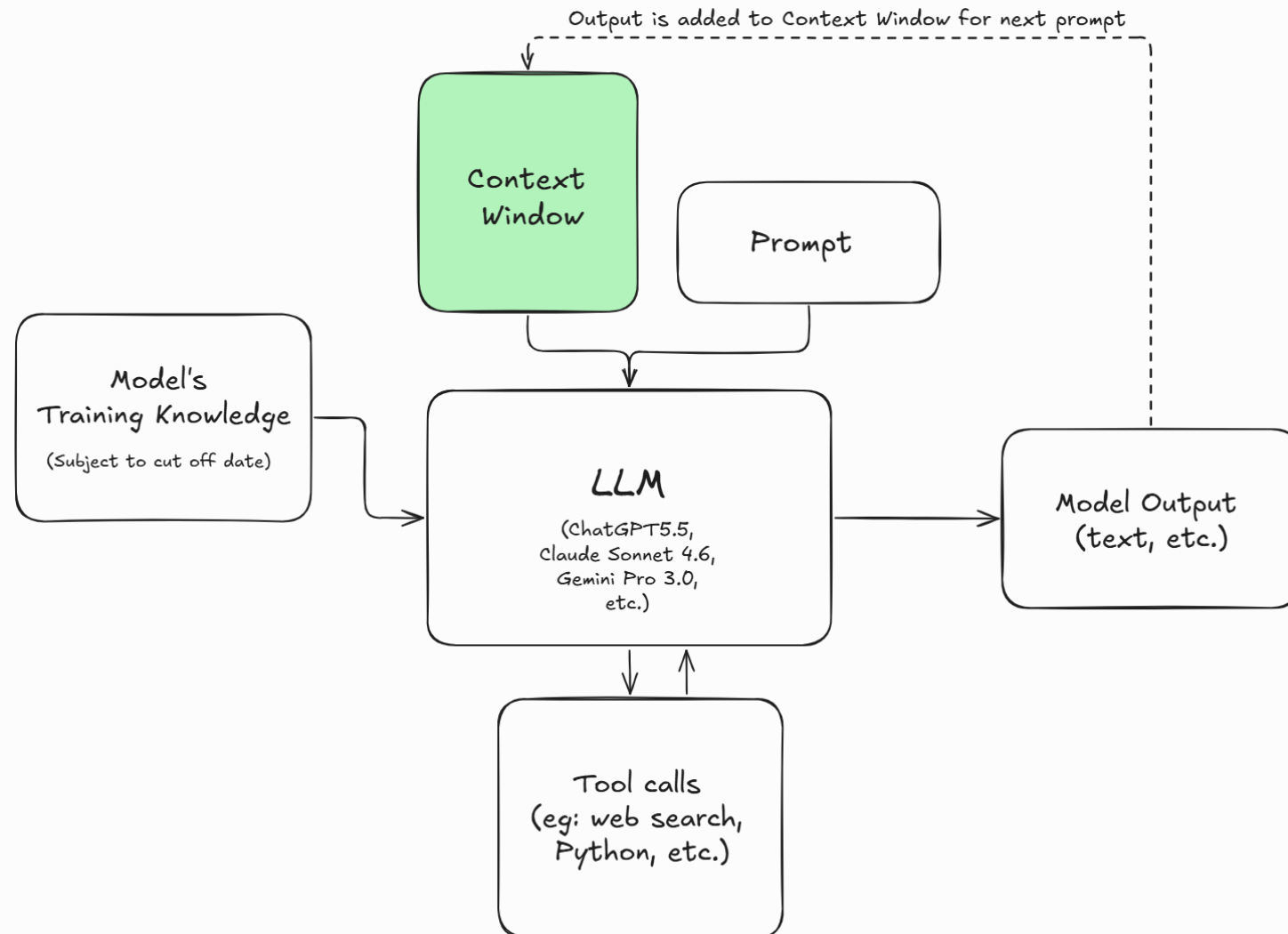
How LLMs Work



Model Knowledge:

- LLMs are training on an enormous amount of data: text from across the internet, open-source books, video, and custom datasets created to train for specific tasks.
- However, LLM memory is not like a database; it's *lossy*. The more frequently data occurs in the training set, the more likely the LLM will "remember" it.
- Subject to a **cut off date**: information is only current up to the date the training dataset created.
- Model knowledge is static and does not improve over time; only increases when new models are released.

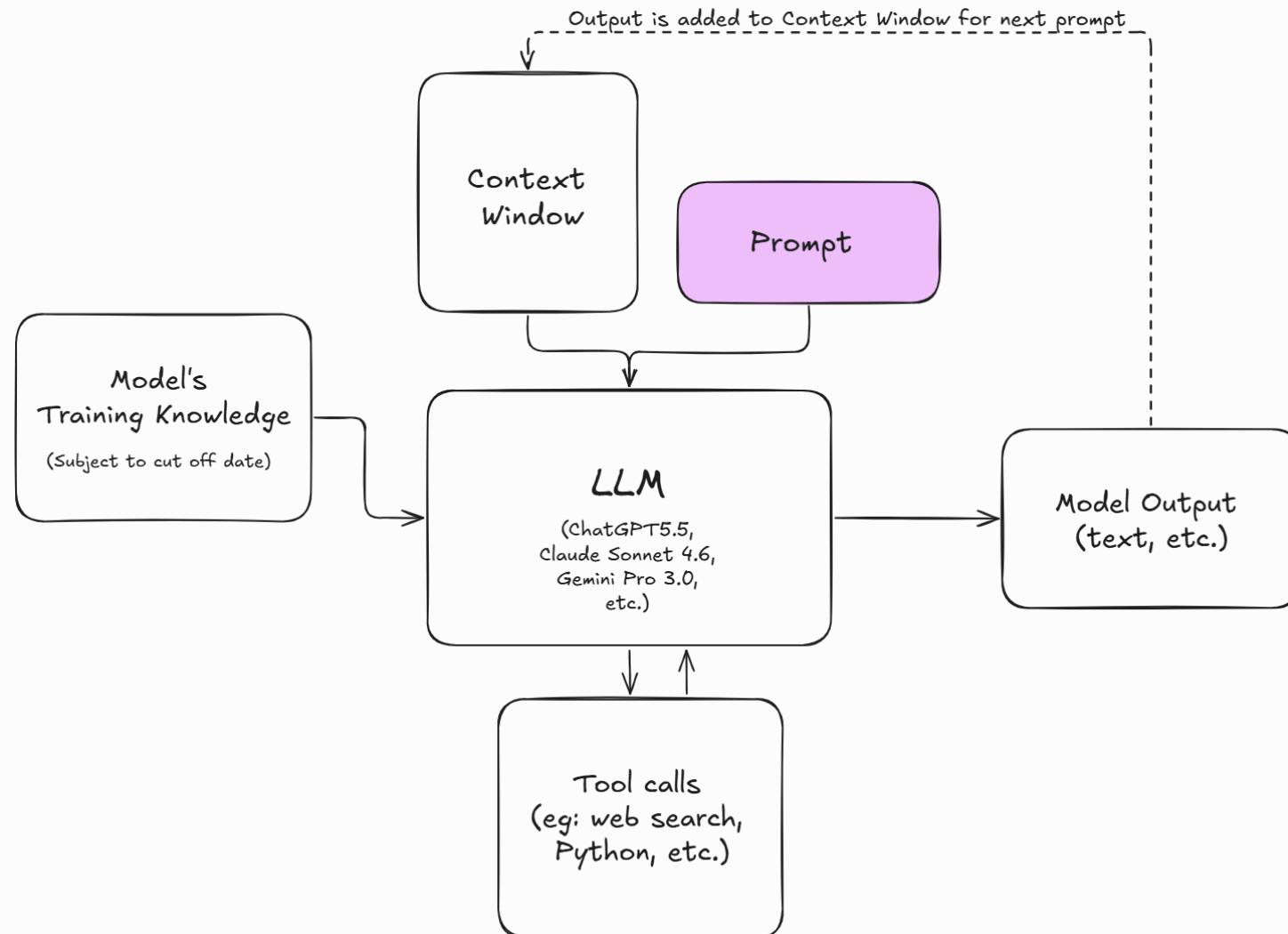
How LLMs Work



Context Window:

- The context window is a key concept for managing LLMs.
- For a conversation, the context window combines all files or data the user uploaded to the LLM, plus all the messages of the conversation so far.
- Information in the context window is given a much higher weight by the LLM than the model's training memory.
- However, subject to significant constraints: most models restrict size to 200,000 tokens; model performance degrades as the context window fills up ("context rot").

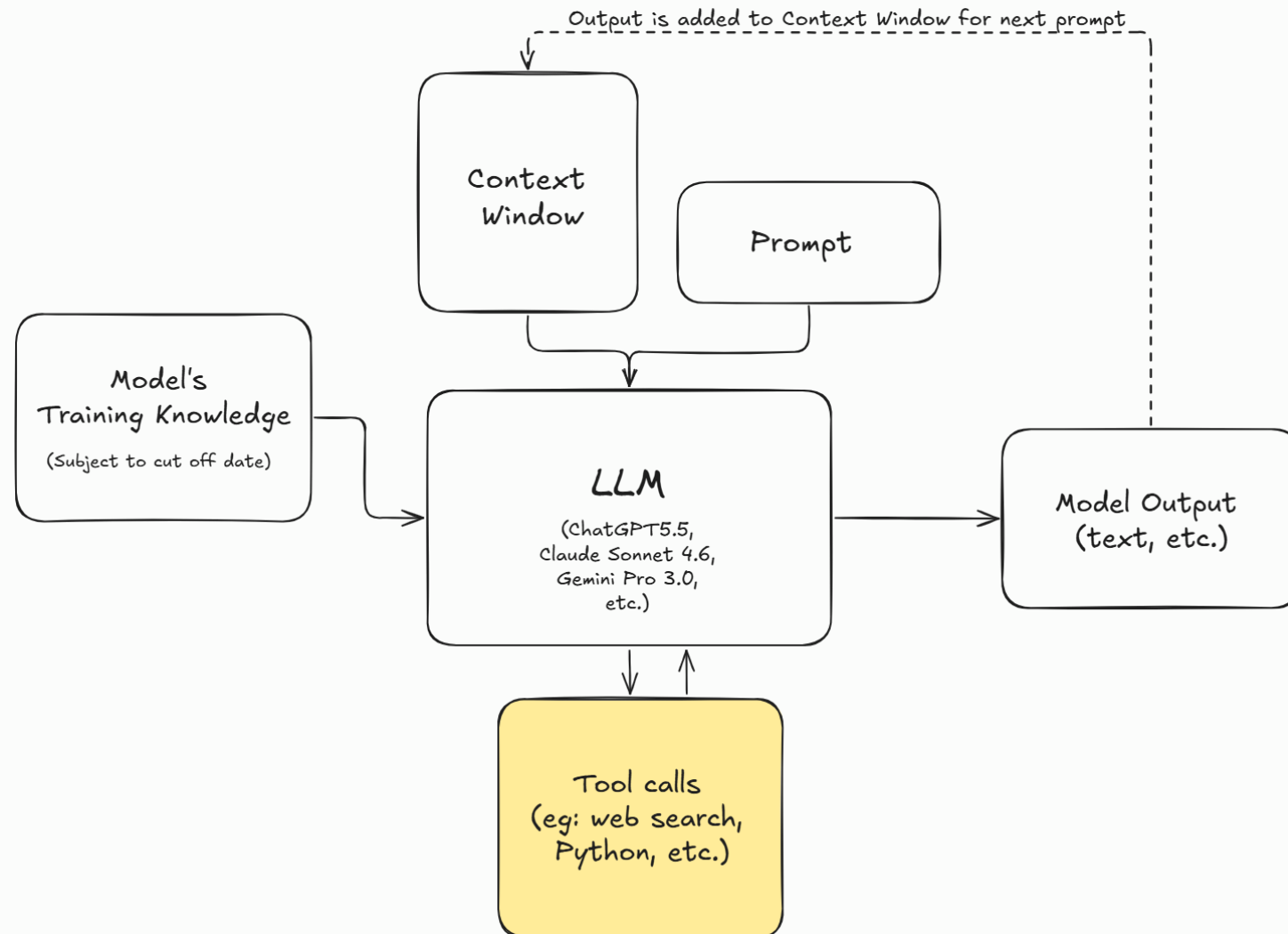
How LLMs Work



Prompts:

- This is the interface where the user provides data and instructions to the LLM.
- Information from the context window is combined with the latest prompt and passed to the model.
 - The output is then added to the context window for the next prompt.
- A well-written prompt with examples and clear instructions can produce a much higher quality output than a simple prompt.

How LLMs Work



Tool Calls:

- LLMs have now been trained to call tools to answer queries they cannot solve with their own knowledge.
- For example, if a model decides it doesn't know the answer to a question, it can conduct a web search to gather additional information.
- LLMs can call tools to perform coding tasks (eg: Python), process PDFs, search emails or shared drives (if enabled). Third party developers can create custom tools (MCP) to perform additional tasks.

LLMs 101: Advanced Features

Thinking Models and other GenAI models

"Thinking" Models:

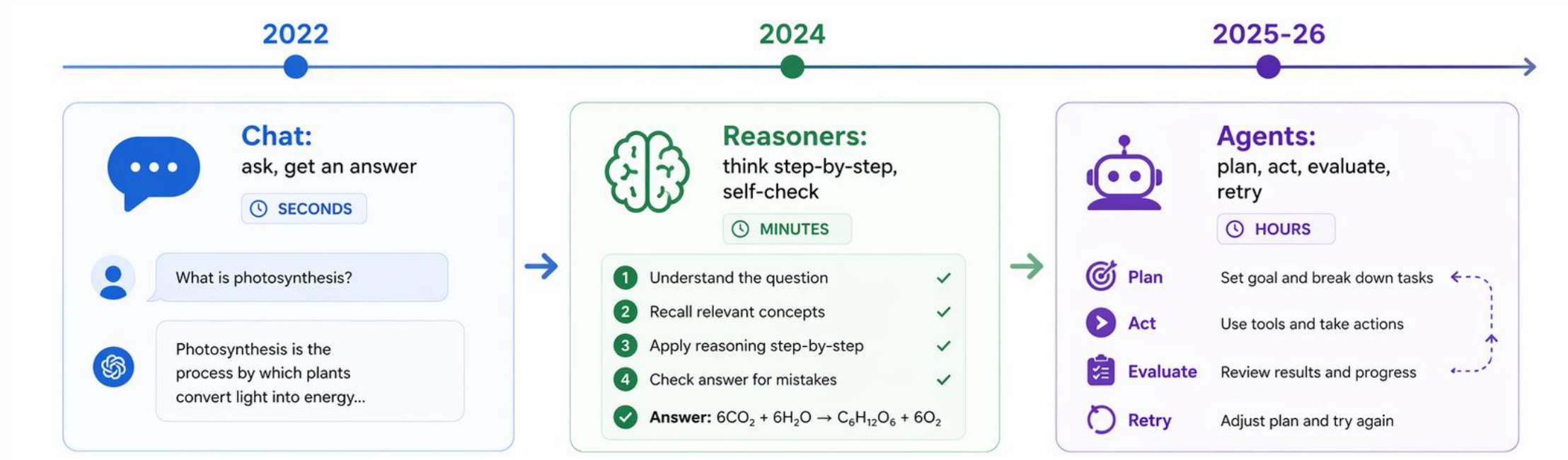
- Models that show their reasoning process: emit intermediate tokens to break down problems "step-by-step":
 - Can be toggled on or off for ChatGPT 5.5 and Claude Opus 4.7; Gemini 3.1 Pro vs Flash
- Take more time to "think through" complex problems
- Better at multi-step reasoning and complex analysis
- Consume more tokens, so can reach usage limits faster
- **Use thinking models for:** Complex problem-solving, multi-step analysis, code debugging, strategic planning
- **Use standard models for:** Quick responses, simple tasks, creative writing

Multi-Modal Input:

- Frontier LLMs can also take in image or audio input and can directly output audio.
- Special tokenizers are used to decode the non-text data and feed it to the LLM.
- Concepts like model knowledge, context window and prompting still apply.
- Other GenAI tools like image, audio and video generation combine LLMs with very different diffusion model architecture.

LLMs 101: From Chatbots to Agents

Agents emerge as a dominate technology in 2026...



- AI agents can now execute multi-step tasks autonomously: research, write, analyze, and take action
- Example: engineers are shifting from *writing* code to *coordinating agents* that write code. The same pattern is emerging across knowledge work.

LLMs 101: Strengths & Limitations

What are LLMs good at and where do they struggle?

Strengths:

- **Language tasks:** Writing, editing, summarizing, translating
- **Pattern recognition:** Identifying themes, categorizing information
- **Code generation:** Writing and debugging code in many languages
- **Creative work:** Brainstorming, ideation, drafting
- **Explanation:** Breaking down complex topics
- **Format transformation:** Converting between document types

Limitations:

- **Precise calculation / counting:** Not a calculator (though getting better)
- **Perfect factual accuracy:** Can "hallucinate" information
- **Real-time data:** Knowledge cutoff dates (usually several months old)
- **Consistent logic:** Can make reasoning errors
- **Accessing private data:** Cannot access information not in training data or context

LLMs 101: Hallucinations & Confidence

How can we be confident in an LLM's answer?

- Hallucinations occur when an LLM generates plausible-sounding but incorrect information.
- The model "fills in gaps" with invented details, presented with the same confidence as factual information
- Prevalence is going down, as models have been trained to be less sure about their own answers and also supplement knowledge with web searches.

Hallucination Red Flags:

- Extreme specificity about obscure topics
- Citations that seem "too perfect" for your query
- Information you can't verify elsewhere
- Details that contradict known facts
- Overly confident answers to ambiguous questions

How to deal with hallucinations:

- ✓ Always review and verify outputs
- ✓ Ask for sources and citations
- ✓ Use multiple models for important decisions ("council of advisors")
- ✓ Cross-reference with authoritative sources
- ✓ Trust more for formatting and structure, less for specific facts
- ✓ Ask how rare is this information across the internet?
- ✓ The less you know about a topic, the more cautiously you should proceed

Troubleshooting with LLMs

LLMs are one of the greatest learning tools ever created



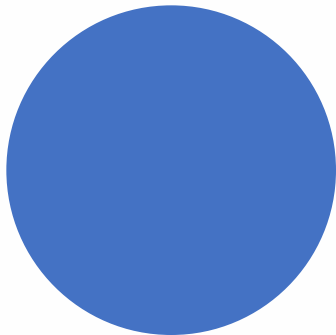
- LLMs are powerful learning tools that can help you learn new skills and troubleshoot problems
 - ▶ including learning how to use LLMs!
- Ask the model for help:
 - It can explain concepts at any level
 - Iterate and break down answer with follow up questions
 - “What should I know about X?”

Troubleshooting techniques:

- Paste screenshots of error messages
- Ask "What should I do next?"
- Request step-by-step explanations
- Have it explain like you're new to the topic



Prompt and Context Engineering



The Art of Prompting

Core techniques for prompting LLMs

- **Key Idea:** Your prompt + context window = How you control the LLM
- Shift from asking questions to **giving instructions**:
 - ❌ "Tell me about renewable energy"
 - ✅ "Create a 2-page briefing on renewable energy policy trends in Canada for provincial policymakers, focusing on grid integration challenges"
- **The difference:** Specificity, context, and clear expectations
- Use "thinking models" when instructions are complex or multi-step.
- Finally, what context you give the model is crucial. We will loop back to this shortly.

Core Prompting Pattern

1. **ROLE/CONTEXT** → Who should the AI be?
 - "You are an expert energy policy analyst"
2. **TASK** → What do you want done?
 - "Summarize this report"
3. **FORMAT** → What should the output look like?
 - "Create a 2-page briefing with executive summary"
4. **CONSTRAINTS** → What are the boundaries?
 - "Use Canadian examples only; target audience is provincial policymakers"
5. **EXAMPLES** → Show what you want (few-shot learning)
 - Provide a sample of desired output style

The Art of Prompting

Advanced Prompting Techniques

- Use XML-like tags to organize complex prompts:
 - Creates clear structure for the LLM
 - Separates different types of information and reduce ambiguity.
- Meta-Prompting!
 - Ask the LLM to help you craft better prompts
 - It can often structure your request better than your first attempt.

```
<role>You are an energy transition communications expert</role>

<task>Transform this technical report into three outputs</task>

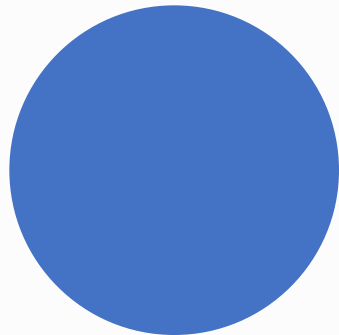
<report>
[Your technical document]
</report>

<outputs>
1. 1-page policy brief
2. Social media thread (5 tweets)
3. Email to stakeholders
</outputs>

<audience>Provincial energy ministers</audience>
```

```
I need to create a prompt that will help me [your goal].
My audience is [audience].
The input will be [type of input].
I want the output to [describe output].

Can you help me write an effective prompt for this task?
```



Managing the Context Window

Context is critical for performance

Key idea: Providing context (information) is critical to LLM performance

- Without context: AI falls back on generic training data and outputs are general and less useful
- With good context: LLMs understand your specific needs and outputs are tailored and actionable

Users provide context by uploading files, pasting information (text) into the prompt window, or connecting to external data sources like email or cloud storage.

LLMs can also replicate *styles* and use other documents as *templates*, so context can also inform the *form* of the output.

Note: Make sure you are following organizational data use guidelines before uploading any sensitive, personal or confidential data to an LLM.

Data + Form = Effective Output

1. DATA/INFORMATION Context:

- Background documents
- Previous research
- Organizational policies
- Stakeholder information

2. FORM/TEMPLATE Context:

- Document structures you use
- Writing style guides
- Format requirements
- Example outputs

Combining both produces the best results

Managing the Context Window

Techniques for managing the context window

- Be selective about what you put in the context window; more ≠ better
 - Context windows have hard limits and performance degrades as they fill up
- Unnecessary or low-quality documents can confuse the LLM

Guidelines:

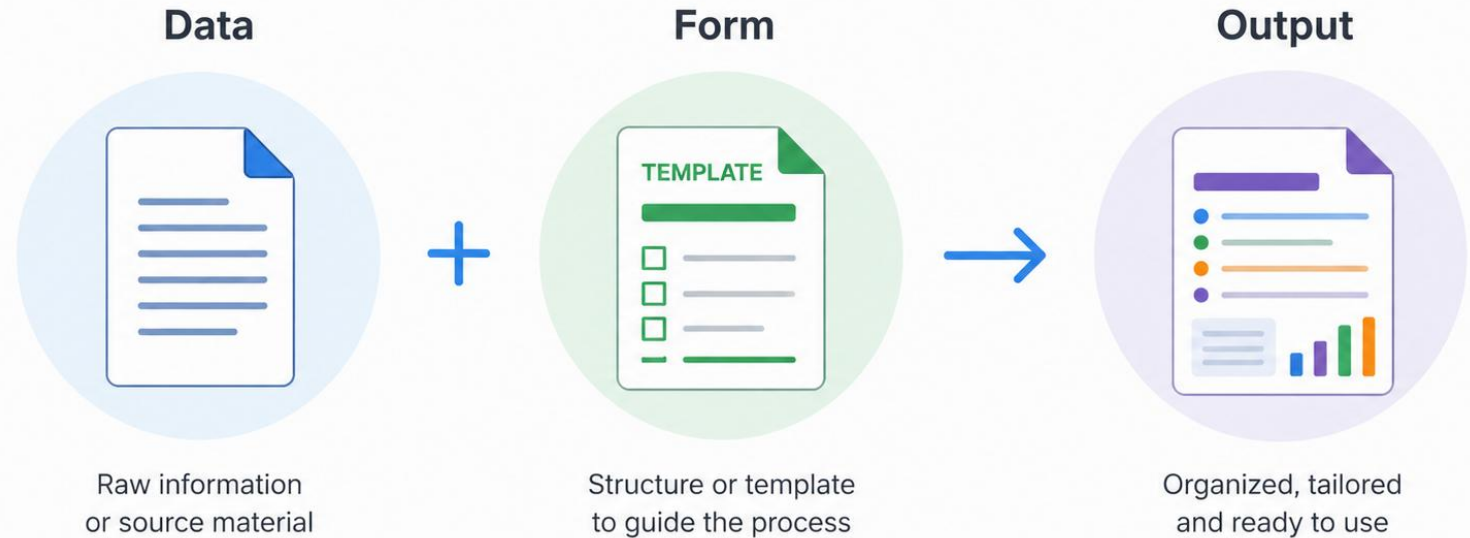
-  Only provide context that is necessary
-  Don't dump "all the files" into the window
-  Higher quality, relevant context → higher quality output
-  Curate and organize information before uploading
-  Frequently start a new conversation to clear the context window
-  Don't mix conversations: open separate tabs and use multiple chat sessions



Prompting + Context

Putting it all together

- Let's now work through some examples where we can combine what we have learned:
- Example 1: Summarizing a podcast, focusing on particular details: ([link](#))
- Example 2: Creating a new project proposal modelled off an existing proposal



Best Practices

Easy practices to quickly improve your productivity using LLMs

1. Build a Prompt Library

- Create shared templates for common tasks (summaries, social posts, emails, briefings)
- Ensure consistency across team outputs and share what works

2. Create Reusable Context Documents

- Develop organizational reference files (mission, values, terminology, style guides) or project briefing files.
- Reuse documents for consistency and speed

3. Use Projects & Memory Features

- Upload persistent files and set custom instructions for specific workflows
- Organize related work in dedicated spaces (research areas, ongoing initiatives)
- Share context without re-uploading every time

General Insight:

Where should I *allocate my thinking* on a project and where can I rely on an LLM to automate the low-value, “drudge work” side of knowledge work?

Additional Resources

For further reading

Andrej Karpathy:

- “How I use LLMs” ([link](#))
- “Intro to Large Language Models” ([link](#))

Prompting Guides:

- OpenAI – GPT-5.5 Prompting Guide ([link](#))
- OpenAI – Using GPT-5.5 ([link](#))
- Anthropic – Claude Prompting Best Practices ([link](#))
- Google – Gemini 3 Prompting Guide ([link](#))

Learning Academies:

- Anthropic Academy ([link](#))
- OpenAI Academy ([link](#))



Thank You!



Sean Mullin

sean.mullin@gmail.com

www.seanmullin.com

